

PROJECT FORMULATION

Concept of Project Formulation

Project formulation is the systematic development of a project idea for the eventual objective of arriving at an investment decision. It involves a step-by-step investigation and development of project idea. And it provides a controlled mechanism for restricting expenditure on project development.

According to **B.B. Goel**, Project formulation refers to a series of steps to be taken to convert an idea or aspiration into a feasible plan of action.

According **G. Myrdal**, Project formulation is one of the basic techniques through which planning can be changed from an idea to an institutional and rational base.

Project formulation is a process involving the joint efforts of a team of experts. Each member of the team should be familiar with the broad strategy, objectives and other ingredients of the project. The government official who deals with the project's final clearance has to be treated as forming part of the team. A well-formulated feasibility report provides a medium, which cuts across scientific, social and positional prejudices and provides a common meeting ground for all those who have a contribution to make in successful implementation of a project.

Project team should consist of experts in major substantive fields of the project. Depending on the situation any large project should comprise the following team members.

- a. One industrial economist
- b. One market analyst
- c. One or more technologist/engineer specializing in the appropriate industry.
- d. One mechanical and/or industrial engineer.
- e. One management accounting expert.

Stages of Project Formulation

1. Project identification

Project identification is concerned with the collection, compilation and analysis of economic data for the eventual purpose of locating possible opportunities for investment and with the development of the characteristics of such opportunities.

2. Technical Analysis

Technical analysis includes the following:

1. Input analysis

a. Site: Location of enterprise; owned or leasehold land; industrial area; No Objection Certificate from the Municipal Authorities if the enterprise location falls in the residential area.

- b. Raw Material:** Requirement of raw material, whether inland or imported, sources of raw material supply.
- c. Human Resource:** Availability of skilled, Semi-skilled labor in the area, arrangements for training laborers in various skills.
- d. Technology Selected:** The selection of technology, arrangements made for acquiring it should be mentioned in the business plan.
- e. Utilities:** These include: **Power:** Requirement for power, load sanctioned, availability of power. **Fuel:** Requirement for fuel items such as coal, coke, oil or gas, state of their availability. **Water:** The sources and quality of water should be clearly stated in the project report.
- f. Pollution Control:** The aspects like scope of dumps; sewage system and sewage treatment plant should be clearly stated in case of industries producing emissions.
- g. Communication System:** Availability of communication facilities, e.g., telephone, fax, internet facility, etc. should be stated in the project report.
- h. Transport Facilities:** Requirements for transport, mode of transport, potential means of transport, distances to be covered, bottlenecks etc., should be stated in the business plan.
- i. Other Common Facilities:** Availability of common facilities like machine shops, welding shops and electrical repair shops etc. should be stated in the report.
- j. Production Process:** A mention should be made for process involved in production and period of conversion from raw material into finished goods.
- k. Machinery and Equipment:** A complete list of items of machinery and equipments required indicating their size, type, cost and sources of their supply should be enclosed with the project report.
- l. Capacity of the Plant:** The installed licensed capacity of the plant along with the shifts should also be mentioned in the project report.
- m. Research and Development:** A mention should be made in the project report regarding proposed research and development activities to be undertaken in future.

2. Demand and Supply analysis

State the total expected demand for the product and present supply position. This should also be mentioned how much of the gap will be filled up by the proposed unit. We can determine the demand of consumer's in following way

- a. Survey of buyer intentions;
- b. Composite of sales force opinion;
- c. Expert's opinion;
- d. Market test method;
- e. Statistical method.

3. Feasibility study

Feasibility studies aim to objectively and rationally uncover the strengths and weaknesses of an existing business or proposed venture, opportunities and threats as presented by the environment, the resources required to carry through, and ultimately the prospects for success.

4. Project Appraisal

Project appraisal is the systematic and comprehensive review of the economic, environmental, financial, social, technical and other such aspects of a project to determine if it will meet its objectives.

According to **B B Goel**, Project appraisal is a tool to examine as to whether in the given situation, it would be most realistic, reliable and reasonable one to commit resources or not.

According to **P.K. Motto**, Project appraisal is the process of evaluation the salient features of feasibility analysis, techno-economic analysis, design and network analysis, input analysis, financial analysis and social cost-benefit analysis of a project.

The following specific aspects must be covered

- Technical Viability
- Commercial Viability - Market Demand
- Financial Viability
- Economic Viability
- Environmental Viability
- Managerial Viability

In essence, the project should be technically feasible and adequate demand should exist for selling the products generated by the operation of the project facilities. There should be adequate financial returns, and adequate returns to the national economy. All environmental requirements must be possible to be fulfilled and the enterprise must have the capability to manage the facilities efficiently and effectively after the project is over.

PROJECT PLAN

Definition of Project Plan

Project planning is a procedural step in project management, where required documentation is created to ensure successful project completion. Documentation includes all actions required to define, prepare, integrate and coordinate additional plans. The project plan clearly defines how the project is executed, monitored, controlled and closed.

A Project Plan sets out the phases, activities and tasks needed to deliver a project. The timeframes required to deliver the project, along with the resources and milestones are also shown in the Project Plan.

According to the **Project Management Body of Knowledge**, Project plan is a formal, approved document used to guide both *project execution* and *project control*. The primary uses of the project plan are to document planning assumptions and decisions, facilitate communication among *stakeholders*, and document approved scope, cost, and schedule *baselines*. A project plan may be summarized or detailed.

According to **R L Martino**, A project plan establishes the duration of the project, the resources needed to complete each activity and the required sequence of performance of each job.

Project planning requires an in-depth analysis and structuring of the following activities:

- Setting project goals
- Identifying project deliverables
- Creating project schedules
- Creating supporting plans

Steps of Project Planning

The key to a successful project is in the planning. Creating a project plan is the first thing you should do when undertaking any kind of project. Project planning consists of a different steps these are as follows:

Step 1: Project Goals

A project is successful when the needs of the stakeholders have been met. A stakeholder is anybody directly or indirectly impacted by the project.

As a first step, it is important to identify the stakeholders in your project. It is not always easy to identify the stakeholders of a project, particularly those impacted indirectly. Examples of stakeholders are:

- The project sponsor.
- The customer who receives the deliverables.
- The users of the project outputs.
- The project manager and project team.

Once you understand who the stakeholders are, the next step is to find out their needs. The best way to do this is by conducting stakeholder interviews. Take time during the interviews to draw out the true needs that create real benefits. Often stakeholders will talk about needs that aren't relevant and don't deliver benefits. These can be recorded and set as a low priority.

The next step, once you have conducted all the interviews, and have a comprehensive list of needs is to prioritise them. From the prioritised list, create a set of goals that can be easily measured. A technique for doing this is to review them against the SMART principle. This way it will be easy to know when a goal has been achieved.

Once you have established a clear set of goals, they should be recorded in the project plan. It can be useful to also include the needs and expectations of your stakeholders.

This is the most difficult part of the planning process completed. It's time to move on and look at the project deliverables.

Step 2: Project Deliverables

Using the goals you have defined in step 1, create a list of things the project needs to deliver in order to meet those goals. Specify when and how each item must be delivered.

Add the deliverables to the project plan with an estimated delivery date. More accurate delivery dates will be established during the scheduling phase, which is next.

Step 3: Project Schedule

Create a list of tasks that need to be carried out for each deliverable identified in step 2. For each task identify the following:

- The amount of effort (hours or days) required to complete the task.
- The resource who will carryout the task.

Once you have established the amount of effort for each task, you can workout the effort required for each deliverable, and an accurate delivery date. Update your deliverables section with the more accurate delivery dates.

A common problem discovered at this point, is when a project has an imposed delivery deadline from the sponsor that is not realistic based on your estimates. If you discover this is the case, you must contact the sponsor immediately. The options you have in this situation are:

- Renegotiate the deadline (project delay).
- Employ additional resources (increased cost).
- Reduce the scope of the project (less delivered).

Use the project schedule to justify pursuing one of these options.

Step 4: Supporting Plans

This section deals with plans you should create as part of the planning process. These can be included directly in the plan.

a. Human Resource Plan

Identify by name, the individuals and organisations with a leading role in the project. For each, describe their roles and responsibilities on the project.

Next, describe the number and type of people needed to carryout the project. For each resource detail start dates, estimated duration and the method you will use for obtaining them.

Create a single sheet containing this information.

b. Communications Plan

Create a document showing that needs to be kept informed about the project and how they will receive the information. The most common mechanism is a weekly or monthly progress report, describing how the project is performing, milestones achieved and work planned for the next period.

c. Risk Management Plan

Risk management is an important part of project management. Although often overlooked, it is important to identify as many risks to your project as possible, and be prepared if something bad happens.

Here are some examples of common project risks:

- Time and cost estimates too optimistic.
- Customer review and feedback cycle too slow.
- Unexpected budget cuts.
- Unclear roles and responsibilities.
- Stakeholder input is not sought, or their needs are not properly understood.
- Stakeholders changing requirements after the project has started.
- Stakeholders adding new requirements after the project has started.
- Poor communication resulting in misunderstandings, quality problems and rework.
- Lack of resource commitment.

Risks can be tracked using a simple risk log. Add each risk you have identified to your risk log; write down what you will do in the event it occurs, and what you will do to prevent it from occurring. Review your risk log on a regular basis, adding new risks as they occur during the life of the project. Remember, when risks are ignored they don't go away.

Having followed all the steps above, you should have a good project plan. Remember to update your plan as the project progresses, and measure progress against the plan.

PROJECT SCHEDULE

Definition of Scheduling

In project management, a schedule consists of a list of a project's terminal elements with intended start and finish dates. Terminal elements are the lowest element in a schedule, which is not further subdivided. Those items are often estimated in terms of resource requirements, budget and duration, linked by dependencies and scheduled events.

A schedule is the conversion of a project action plan an operating timetable. It serves as the basic for monitoring and controlling project activity and taken together with the plan and budget.

According to **R L Martino**, A schedule depicts the expected start and finish time of each job. It is produced by allocating resources upto the limit of availability according to the requirements given by the plan.

Scheduling is the determination of time required for execution of each operation and the time order in which each operation has to be carried out, to meet the plan objectives.

Network Techniques (Critical Path Method)

Network techniques Developed in 1950's CPM by DuPont for chemical plants (1957)

Complex project, like the above example, require a series of activities, some of which must be performed sequentially and others that can be performed in parallel with other activities. This collection of series and parallel tasks can be modeled as a network.

CPM models the activities and events of a project as a network. Activities are shown as nodes on the network and events that signify the beginning or ending of activities are shown as arcs or lines between the nodes. The Figure shows an example of a CPM network diagram:

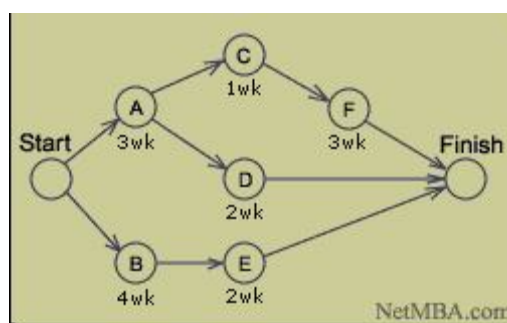


Figure: CPM Diagram

Steps in CPM Project Planning

1. Specify the individual activities

All the activities in the project are listed. This list can be used as the basis for adding sequence and duration information in later steps. (Identify and list all project activities.)

2. Determine the sequence of the activities

Some activities are dependent on the completion of other activities. A list of the immediate predecessors of each activity is useful for constructing the CPM network diagram. (Define dependencies and order of activities (which comes first).

3. Draw the Network Diagram

Once the activities and their sequences have been defined, the CPM diagram can be drawn. CPM originally was developed as an *activity on node* network. (Create the network (usually Activity-on-Node format).

4. Estimate activity completion time

The time required to complete each activity can be estimated using past experience. CPM does not take into account variation in the completion time. (Assign duration to each activity (based on experience).

5. Identify the Critical Path

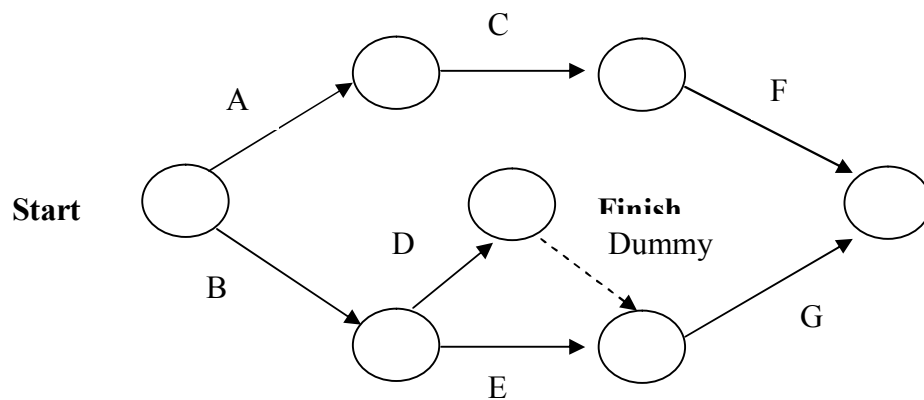
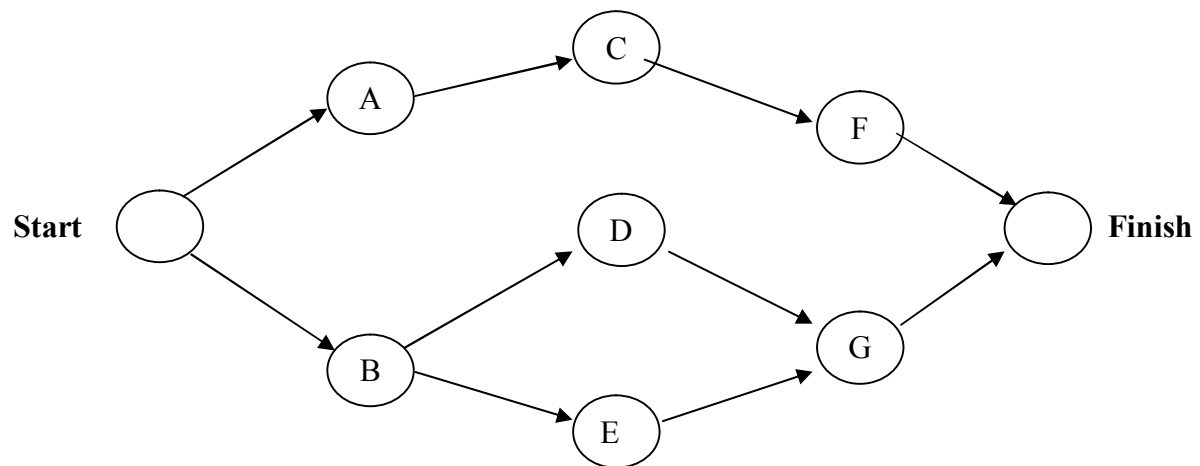
The critical path is the longest-duration path through the network. The significance of the critical path is that the activities that lie on it cannot be delayed without delaying the project. Because of its impact on the entire project, critical path analysis is an important aspect of project planning. (Find the longest path in the network using: ES (Earliest Start); EF (Earliest Finish); LS (Latest Start) and LF (Latest Finish)

6. Update CPM diagram

As the project progresses, the actual task completion times will be known and the network diagram can be updated to include this information. A new critical path may emerge, and structural changes may be made in the network if project requirements change. (Revise the network as the project progresses and times change.)

Arrow and Node diagram

Activity	Immediate predecessor activity
A	-
B	-
C	A
D	B
E	B
F	C
G	D, E

**Figure: Arrow Diagram****Figure: Node Diagram**

Following are the information of a construction project, using the given information- Draw a network diagram and determine critical path.

Activity	Time of work (in week)	Immediate predecessor activity
A	6	-
B	9	A
C	8	A
D	12	A
E	11	B
F	7	C, D
G	12	E, F
H	14	D
I	9	G, H

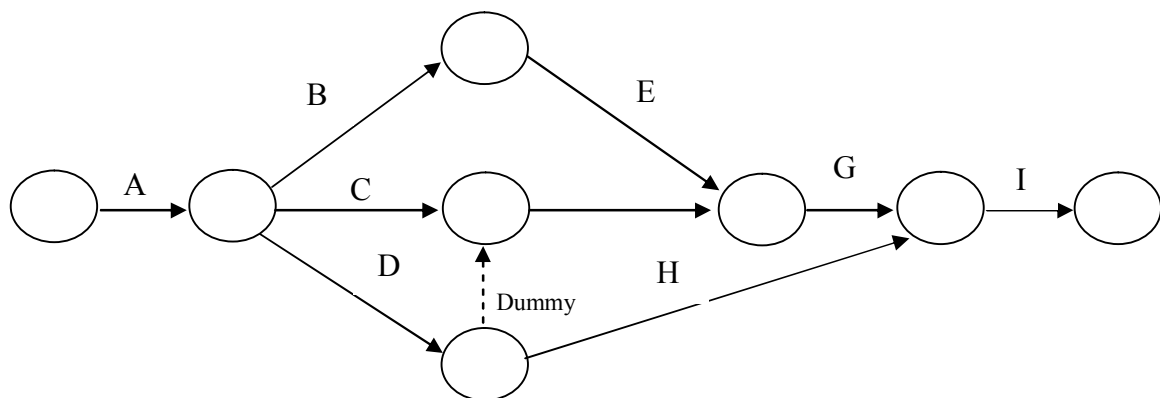


Figure: Network Diagram

Determination of Critical path

If we want to determine the critical path, we have to find out the alternative paths from the above diagram and determine their length. We have found out the following alternative paths from the above diagram:

1. A – B – E – G – I = 6+9+11+12+9 = 47 weeks
2. A – C – F – G – I = 6+8+7+12+9 = 42 weeks
3. A – D – Dummy – F – G – I = 6+12+0+7+12+9 = 46 weeks
4. A – D – H – I = 6+12+14+9 = 41 weeks

We know that, among all the alternative paths, longest path is determined as Critical Path. A-B-E-G-I path within the above mentioned diagram is the longest path among all the alternative paths. Hence, Critical path = 47 weeks.

PROJECT IMPLEMENTATION

Concept of project implementation

Implementation is the stage where all the planned activities are put into action. Before the implementation of a project, the implementors (spearheaded by the project committee or executive) should identify their strength and weaknesses (internal forces), opportunities and threats (external forces). The strength and opportunities are positive forces that should be exploited to efficiently implement a project. The weaknesses and threats are hindrances that can hamper project implementation. The implementors should ensure that they devise means of overcoming them.

Monitoring is important at the implementation phase to ensure that the project is implemented as per the schedule. This is a continuous process that should be put in place before project implementation starts. As such, the monitoring activities should appear on the work plan and should involve all stakeholders. If activities are not going on well, arrangements should be made to identify the problem so that they can be corrected. Monitoring is also important to ensure that activities are implemented as planned. This helps the implementors to measure how well they are achieving their targets. This is based on the understanding that the process through which a project is implemented has a lot of effect on its use, operation and maintenance.

Implementation is the carrying out, execution or practice of a plan, a method or any design for doing something. As such, implementation is the action that must follow any preliminary thinking in order for something to actually happen.

Steps of project implementation

Managing a project is a process. It consists of a series of orderly steps that if carried out properly will ensure effective communication among project personnel and the project's clients and contain enough flexibility to handle the unexpected bumps along the way. In fact, one of the biggest detriments of the success of a project is how effectively the project manager carries out this process of project implementation. The following is a brief guide to the basic steps of project implementation. Keep in mind that all projects regardless of size and scope, will generally follow the same structure.

Step 1: Project managers should seek guidance

First-time project managers should find someone who can guide them through the process of completing the project. Ideally, this adviser should be someone in a similar business since that person may also benefit from the project and thus be more motivated to help. Where this is not possible, the manager should seek counsel from any individual who is experienced in project management in general. Even in this case, however, this consultant should at least be familiar with the industry and/or type of project.

Step 2: Acquire appropriate project management tools and software

In order to facilitate project completion, project managers should ensure that the proper supporting infrastructure is in place. This includes: communications devices and software, computer hardware and any software specific to project management.

Step 3: Clarify the details of the project

One of the first tasks assigned to the project manager is the creation of a project proposal. Included in the proposal should be the following:

- What will the project accomplish or what are its goals?
- What is the time frame for project completion?
- What is the project budget?
- Who will be involved in completing the project along with assigned tasks?
- What method(s) will be used to complete the project?
- How will project uncertainties be dealt with?

Step 4: Communicate job responsibility among project team members

Often, several individuals may be working on a project with each offering their own input. Where there is a team of people assigned to complete the project, then all members should be clear about their assigned roles and duties. There should also be a system in place to ensure that everyone is updated regularly throughout the project schedule.

Step 5: Make sure a system is in place to monitor progress

The project manager must establish a plan to measure the project's progress and efficiency in order to determine whether additional information or resources are needed to complete the project, whether resources should be shifted around, and occasionally whether a new approach to the project should be taken. The main focal points for any project manager include: performance management and quality control, as well as how well the project personnel are adhering to the estimated time frame and monetary budget.

Step 6: Document project information

Providing easy to understand and accurate documentation helps to ensure that all parties involved are clear about the project's goals, the time frame for project completion, the task agenda, and the budget throughout the duration of the project. The manager should also carefully record any project assessments made along the way as well as any problems encountered or changes that were initiated.

Step 7: Communicate with upper management and project clients

Aside from regular meetings with team members, the project manager is also responsible for ensuring that supervisors and/or clients are kept well-informed throughout the project's progress.

In short, with a little planning and effort, even inexperienced project managers can implement a project effectively and greatly increase the chances that the project will be a success by utilizing these steps of project implementation.

Impediments to project implementation

The main problems faced by project manager during implementation are poor monitoring of progress, not handling risks and poor cost management.

1. Poor monitoring of progress: Project managers some times tend to spend most of their time in planning activity and surprisingly very less time in following up whether the implementation is following the plan.

2. Not handling risks: Risks have an uncanny habit of appearing at the least expected time. In spite of the best efforts of a project manager they are bound to happen. Risks need immediate and focused attention. Delay in dealing with risks cause the problem to aggravate and has negative consequences for the project.

3. Poor cost management: A project manager's success is measured by the amount of cost optimization done for a project. Managers frequently do all the cost optimization during the planning stages but fail to follow through during the rest of the stages of the project.

PROJECT MONITORING AND EVALUATION

Concept of Project Monitoring

Once a project has been planned and financial support been secured, the most important part begins - implementation. It is very rare for any project to go exactly according to plan. In fact it is not uncommon for a project to take on a direction and a momentum that was completely unanticipated during planning. Project management now has the important and difficult task of establishing sufficient controls over the project to ensure that it stays on track towards the achievement of its objectives. This is done by **monitoring**, which can be defined as the systematic and continuous collection, analysis and use of information for management control and decision-making. Project monitoring is an integral part of day-to-day management. Its purpose is to provide the information by which management can identify and solve implementation problems, and assess progress in relation to what was originally planned.

Conceptually, "monitoring" means to check and assess the implementation status of a project during the implementation on a regular basis. The system of watching/ monitoring the progress of a project implementation, besides being an important link in the project cycle, helps in the identification/analysis and removal of bottlenecks and expediting action where projects have stalled or fallen behind schedule.

According to **B.B. Goel**, Monitoring involves watching the progress against time, resources and performance schedules during the execution of the project and identifying lagging areas requiring timely attention and action.

According to **M. Thyagarajan**, Monitoring means periodic checking of progress of works against the targets laid down in order to ensure timely completion of the project.

Monitoring is the regular observation and recording of activities taking place in a project or programme. It is a process of routinely gathering information on all aspects of the project.

Concept of Evaluation

Project Evaluation is a selective exercise that attempts to systematically and objectively assess the progress of the project that has been implemented. The Evaluation involves assessments of differing scopes and depths carried out at periodically in response to evolving needs for knowledge and learning during the effort to achieve an outcome. Project evaluations will assess project relevance, performance and substantive contributions to the achievement of the output, as well as to the achievement of national development priorities. The strategic positions and institutional partnerships developed during the project implementation should also be assessed. Evaluation is very important for sustainability and the development of future projects. Evaluations need to be undertaken before a project can be replicated and further developed.

Evaluation can be defined as a periodic assessment of the relevance, efficiency, effectiveness, impact, economic and financial viability, and sustainability of a project in the context of its stated objectives.

The purpose of evaluation is to review the achievements of a project against planned expectations, and to use experience from the project to improve the design of future projects and programmes. Evaluation draws on routine reports produced during implementation and may include additional investigations by external monitors or by specially constituted missions.

According to **Skylark Chada**, Evaluation is an analytical process for systematically and objectively perceiving the relevance, efficiency, effectiveness and impact of activities in the light of their objectives.

Evaluation-

- is a periodic assessment of project (Mid-term and completion)
- assesses the relevance, efficiency, effectiveness, and sustainability of the project in relation to its objectives
- focuses more on results-to-purpose and purpose over all objectives

PROJECT TEAM

Concept of project team

A project team is a team whose members usually belong to different groups, functions and are assigned to activities for the same project. A team can be divided into sub-teams according to need. Usually project teams are only used for a defined period of time. They are disbanded after the project is deemed complete. Due to the nature of the specific formation and disbandment, project teams are usually in organizations.

The project team consists of the full-time and part-time resources assigned to work on the deliverables of the project all of who will help achieve the project objectives. They are responsible for:

- understanding the work to be completed
- planning out the assigned activities in more detail if needed
- completing assigned work within the budget, timeline, and quality expectations
- informing the project manager of issues, scope changes, risk, and quality concerns
- proactively communicating status, and managing expectations

The project team can consist of human resources within one functional organization, or it can consist of members from many different functional organizations. A cross-functional team has members from multiple organizations. Having a cross-functional team is usually a sign your organization is using matrix management.

PROJECT COST ESTIMATION

Concept of Cost Estimation

Estimating is a critical component of a Project Management process or methodology. Estimates are used for budgeting, planning, forecasting, resource planning, and staffing. Customers also use estimates to make decisions for changing their business and systems. Cost estimates are quantitative assessments of the likely costs of the resources required to complete project activities. They may be presented in summary or in detail. Costs must be estimated for all resources that will be charged to the project. This includes, but is not limited to, labor, materials, supplies, and special categories such as an inflation allowance or cost reserve.

Cost estimating involves developing an approximation (estimate) of the costs of the resources needed to complete projects activities.

An approximation of the probable [cost](#) of a [project](#) computed on the basis of available [information](#). Cost estimates are generally prepared as accurately as possible, to prevent misallocation of [resources](#) or [negative perceptions](#) from [clients](#), [managers](#), or [potential customers](#).

As defined by the *Project Management Body of Knowledge (PMBOK)*, cost estimation is the iterative process of developing an approximation of the monetary resources needed to complete project activities. Project teams should estimate costs for all resources that will be charged to the project. This includes but is not limited to: Human resources, Materials, Equipment, Services, Software, Hardware, Facilities.

SOCIAL COST BENEFIT ANALYSIS

Concept of social cost benefit analysis

The economic evaluation of any investment proposal involves an analysis of social profitability arising out of the investment. Apart from the direct financial benefits as a result of revenue which will flow in the national exchequer from duties and taxes levied, there are various indirect benefits which will accrue such as opening up of employment opportunities in downstream/ancillary industries, transport and other secondary and tertiary sectors of commercial activity. This analysis which basically determines whether an investment is worthwhile from the point of view of the society as a whole, involves adjustment on cost and return to the enterprise taking into account the direct and indirect benefits leading to revalued inputs and outputs.

According to **P.K. Motto**, Social cost benefit analysis is the process of evaluating a project from the point of view of the total impact which will have on the economy of the nation.

Social cost benefit analysis (SCBA) is a methodology for evaluating project from social point of view. It is also referred as economic analysis and is developed for evaluating investment project from the point of view of the society. In recent years particularly in the developing countries where GOVT is playing significant role in economic development (SCBA) is playing a very important role. It is concerned with tactical decision making within the framework of broad strategic choice defined by planning at the macro level.